

COURSE INFORMATION

Description:

Learners develop foundational web development skills using HTML, CSS, and JavaScript to create structured, styled, and interactive web pages. The course emphasizes building multi-page websites, applying design principles for usability and readability, and implementing basic interactivity through JavaScript. Learners practice debugging and refining web applications using progressive techniques and are introduced to AI-assisted tools to support development. By the end of the course, students are able to design, build, and explain functional web solutions.

Career Cluster: Information Technology

Instructional Level: A.A.S. - Associate in Applied Science

Total Credits: 3

Total Hours: 72

TEXTBOOKS

Responsive Web Design with HTML5 and CSS: Build future-proof responsive websites using the latest HTML5 and CSS techniques - 5th ed. Edition - Ben

JavaScript and jQuery: Interactive Front-End Web Development - 1st Edition - Jon Duckett

HTML 5: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

CSS - Cascading Style Sheets: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

COURSE COMPETENCIES

1. Structure web content using semantic HTML

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

- 1.1. Multi-page website assignments
- 1.2. HTML structure checkpoints
- 1.3. Project assignments and quiz

Criteria

- 1.1. Create structured web pages using semantic HTML elements
- 1.2. Organize content logically using headings, paragraphs, lists, and links
- 1.3. Build multi-page website structures
- 1.4. Apply consistent page organization
- 1.5. Ensure page structure supports styling, navigation, accessibility, and interactivity

Learning Objectives

- 1.a. Define HTML structure and purpose
- 1.b. Identify common HTML elements
- 1.c. Explain how page structure supports styling and interaction
- 1.d. Describe how multi-page sites organize information

2. Design and style web pages using CSS to create readable, usable, and responsive layouts

Domain: Cognitive Level: Creating Status: WIP

Assessment Strategies

- 2.1. CSS styling assignments
- 2.2. Layout and responsive design tasks
- 2.3. Project assignments and quiz

Criteria

- 2.1. Apply CSS to control layout, spacing, typography, color, and visual hierarchy
- 2.2. Use selectors, classes, and IDs effectively
- 2.3. Apply box model, positioning, flexbox, grid, or other layout techniques as appropriate
- 2.4. Create visually consistent and readable designs
- 2.5. Apply responsive design techniques that support usability across screen sizes

Learning Objectives

- 2.a. Define CSS styling principles
- 2.b. Explain layout concepts such as spacing, sizing, and the box model
- 2.c. Identify responsive design basics
- 2.d. Describe how styling choices affect usability and readability

3. Implement interactive behavior using JavaScript to respond to user input and manipulate web page content

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

- 3.1. JavaScript behavior assignments
- 3.2. DOM interaction exercises
- 3.3. Project assignments and quiz

Criteria

- 3.1. Write basic JavaScript using variables, conditionals, functions and events
- 3.2. Implement interactive behaviors that respond to user input
- 3.3. Connect program logic to visible web page outcomes
- 3.4. Manipulate page content, attributes, or styles dynamically
- 3.5. Explain how JavaScript works with HTML and CSS in a web page

Learning Objectives

- 3.a. Define JavaScript as a programming language (for web behavior)
- 3.b. Explain variables, functions, control flow, and events
- 3.c. Describe how JavaScript enables interactivity
- 3.d. Explain how JavaScript can affect page content and presentation

4. Debug and evaluate web applications using progressive techniques across HTML, CSS, and JavaScript

Domain: Cognitive Level: Analyzing Status: WIP

Assessment Strategies

- 4.1. Debugging exercises
- 4.2. Browser tool practice
- 4.3. Project assignments and quiz

Criteria

- 4.1. Identify and correct layout and styling issues
- 4.2. Identify and correct CSS layout or styling issues
- 4.3. Use console logging to trace JavaScript behavior
- 4.4. Use browser developer tools to inspect and troubleshoot web pages
- 4.5. Diagnose issues that emerge from interactions among HTML, CSS, and JavaScript

Learning Objectives

- 4.a. Define debugging and troubleshooting
- 4.b. Identify common HTML, CSS, and JavaScript issues
- 4.c. Explain debugging strategies
- 4.d. Describe how issues can originate across multiple layers

5. Develop web solutions using structured problem-solving, appropriate tools, and iterative improvement

Domain: Cognitive Level: Creating Status: WIP

Assessment Strategies

- 5.1. Plan and build web pages based on defined requirements
- 5.2. Organize files, folders, page structure, styles, and scripts coherently
- 5.3. Use reusable patterns to reduce repetition and improve maintainability
- 5.4. Improve work through testing, feedback, and revision
- 5.5. Refine solutions for clarity, usability, and correctness

Criteria

- 5.1. Plan and build web pages based on defined requirements
- 5.2. Evaluate and refine solutions for usability and clarity
- 5.3. Iteratively improve solutions based on testing and feedback

Learning Objectives

- 5.a. Describe structured development approaches
- 5.b. Explain how organization supports maintainable web work
- 5.c. Identify strategies for iterative improvement
- 5.d. Describe how planning, building, testing, and refining form a development cycle

6. Use AI-assisted web development responsibly and transparently

Domain: Cognitive Level: Evaluating Status: WIP

Assessment Strategies

- 6.1. Project assignments and quiz

Criteria

- 6.1. Use AI tools within assignment boundaries and course phase expectations
- 6.2. Avoid using AI to bypass required foundational practice
- 6.3. Evaluate AI-generated or AI-suggested HTML, CSS, and JavaScript before using it
- 6.4. Adapt, reject, or revise AI output based on project requirements and observed behavior
- 6.5. Explain what AI assisted with, what changed, and what decisions remained their own

Learning Objectives

- 6.a. Describe appropriate AI use in staged web development learning
- 6.b. Identify risks of relying on AI before foundational understanding is developed
- 6.c. Explain how responsible AI use requires validation, revision, and accountability
- 6.d. Describe how AI can support troubleshooting, refinement, and extension without replacing understanding

7. Solve web development problems using structured strategies

Domain: Cognitive Level: Analyzing Status: WIP

Assessment Strategies

- 7.1. Debugging activities
- 7.2. Iterative web project work
- 7.3. Project assignments and quiz

Criteria

- 7.1. Identify the source of a web development problem
- 7.2. Apply structured troubleshooting before requesting direct answers
- 7.3. Use documentation, examples, browser tools, or targeted support to recover
- 7.4. Compare possible solutions before selecting an approach
- 7.5. Persist through revision when a page, style, or behavior does not work as intended

Learning Objectives

- 7.a. Describe problem-solving strategies for web development
- 7.b. Explain how debugging supports independent recovery
- 7.c. Identify resources and tools that support technical problem solving

8. Communicate web project decisions and technical reasoning clearly

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

- 8.1. Project explanation activities
- 8.2. Code walkthroughs
- 8.3. Presentations or discussions
- 8.4. Project assignments and quiz

Criteria

- 8.1. Explain the structure, styling, and behavior of their web work
- 8.2. Describe design, layout, and interaction decisions clearly
- 8.3. Ask clarifying questions when requirements or expectations are unclear
- 8.4. Document project choices, AI use, or revisions when required
- 8.5. Communicate web project outcomes using appropriate technical vocabulary

Learning Objectives

- 8.a. Explain the role of communication in web development work
- 8.b. Identify ways to describe HTML, CSS, and JavaScript decisions
- 8.c. Describe how clear explanation supports technical accountability

9. Work productively through web project milestones and lab time

Domain: Affective Level: Responding Status: WIP

Assessment Strategies

- 9.1. Project milestones
- 9.2. Lab-based development work
- 9.3. Assignment submission patterns
- 9.4. Project assignments and quiz

Criteria

- 9.1. Use class or lab time to make visible progress on web development tasks
- 9.2. Organize files, notes, assets, and project materials
- 9.3. Follow directions and project requirements
- 9.4. Complete required tasks by stated deadlines
- 9.5. Demonstrate self-direction when beginning, continuing, or refining work

Learning Objectives

- 9.a. Identify productive work habits for web development projects
- 9.b. Explain how organization supports project completion
- 9.c. Describe how sustained iteration contributes to stronger web work

10. Apply feedback, resources, and revision to improve web development learning

Domain: Affective Level: Valuing Status: WIP

Assessment Strategies

- 10.1. Revised project submissions
- 10.2. Reflection or checkpoint activities
- 10.3. Resource-use activities
- 10.4. Project assignments and quiz

Criteria

- 10.1. Use instructor feedback to improve web page quality
- 10.2. Locate and apply appropriate references, examples, or documentation
- 10.3. Revise HTML, CSS, JavaScript, explanations, or project decisions based on evidence
- 10.4. Ask useful questions that support deeper understanding
- 10.5. Demonstrate growth across repeated web development tasks

Learning Objectives

- 10.a. Explain how feedback supports learning in web development
- 10.b. Identify resources that can improve technical understanding
- 10.c. Describe how revision strengthens both project quality and reasoning

11. Collaborate respectfully in web development and technical learning activities

Domain: Affective Level: Responding Status: WIP

Assessment Strategies

- 11.1. Pair programming or driver/navigator activities
- 11.2. Peer support activities
- 11.3. Discussion or presentation
- 11.4. Project assignments and quiz

Criteria

- 11.1. Participate respectfully in technical learning activities
- 11.2. Contribute ideas, effort, or support during collaborative work
- 11.3. Use driver/navigator or similar roles appropriately when assigned
- 11.4. Seek help appropriately when blocked
- 11.5. Support peers without replacing their learning or work

Learning Objectives

- 11.a. Describe expectations for respectful technical collaboration
- 11.b. Explain how peer support can strengthen learning
- 11.c. Identify appropriate boundaries for collaboration in web development work

12. Act professionally in web development coursework and tool use

Domain: Affective Level: Valuing Status: WIP

Assessment Strategies

- 12.1. Lab participation
- 12.2. Project presentations
- 12.3. AI-use documentation
- 12.4. Project assignments and quiz

Criteria

- 12.1. Take responsibility for their work, conduct, and tool use
- 12.2. Follow course, assignment, and technical standards
- 12.3. Use AI, examples, assets, and external resources ethically
- 12.4. Respect people, property, data, and learning environments
- 12.5. Demonstrate readiness, reliability, engagement, and accountability in technical tasks

Learning Objectives

- 12.a. Explain professional expectations in web development coursework
- 12.b. Identify ethical considerations in AI-assisted and resource-supported technical work
- 12.c. Describe how accountability supports trust in software and web development